Nifty crash insurance, tested on real NSE option prices

Subham · github.com/Subham1709/Nifty_Hedge_Real_Data · app: subham-nifty-hedge.streamlit.app

Question. A course project (FDRM, IIT Jodhpur) found that a Nifty 50 portfolio did better with crash insurance bought only when India VIX was low: a 3-month put 5% below the market, bought on the first trading day of a month if VIX was below 13, half sold when VIX reached 22 and the rest as the spike faded. The project had no historical option prices, so every premium came from Black-Scholes with India VIX plus an assumed put skew. Here the same backtest is rebuilt on **1,224,590 real Nifty option prices** from NSE's daily F&O bhavcopies, 2020-01-02 to 2026-04-13.

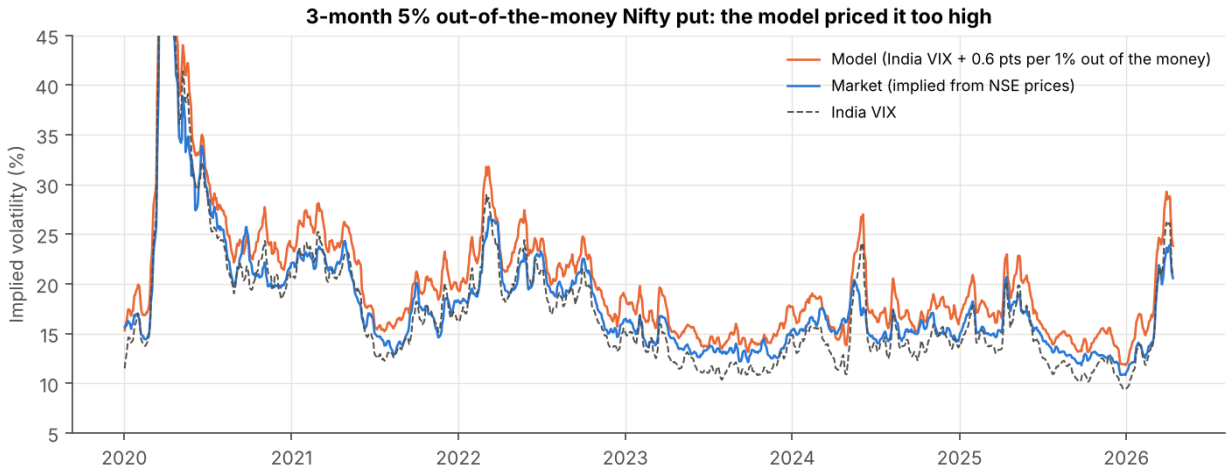
Results

The model priced the insurance too high. Against the market, its premium was a median 33% too high for 3-month puts and 21% for 1-month puts, and higher in 91% of purchases. The market charged an implied volatility of about VIX + 1 point for the 3-month put, not the VIX + 3 the model assumed. The gap was widest when VIX was low, which is when the rule buys.

The rule held up. On real prices it returned **14.8% a year** against 11.1% unhedged, with a worst fall of **-22.8%** against -38.4% (model prices: 13.9%). Every hedge looks cheaper on real prices, but the ranking is unchanged: buying a put every month still cost more than it saved, and buying only when VIX was low still beat no hedge. Holding the VIX-timed put to expiry did even better than the hybrid selling rule, as in the course project.

Strategy	Return a year, real prices	Model prices (course project)	Worst fall, real	Insurance cost a year, real	Puts bought
Unhedged	11.1%	11.1%	-38.4%	0.0%	0
Always 1M 95% put	10.5%	8.3%	-20.9%	0.9%	76
Always 1M 95/85 put spread	10.2%	7.3%	-30.4%	0.9%	76
Always 3M 95% put, rolled monthly	8.6%	6.2%	-15.7%	2.8%	76
VIX-timed 3M put, hold to expiry	17.5%	17.2%	-17.2%	-4.9%	22
VIX-timed 3M put, hybrid selling (proposed)	14.8%	13.9%	-22.8%	-3.0%	22
VIX-timed 3M put, sell all at VIX 22	11.8%	11.0%	-36.2%	-0.6%	22

Jan 2020 to Apr 2026. Real prices include a 2% bid/ask haircut on every trade. A negative insurance cost means the puts earned more than they cost.



Robustness

The result survives bid/ask haircuts from 0% to 10% (14.9% to 14.5% a year), liquidity cut-offs from 10 to 200 contracts, and buy thresholds from VIX 12 to 14. A 1-month or 2-month put does not work: the January 2020 put expired before the COVID crash. From July 2020 the rule returned 16.1% against 15.4% unhedged, so most of the full-period lead comes from one put bought in January 2020.



Method

Contracts. Each put is a real listed contract: the monthly or quarterly expiry nearest the chosen length (80 to 91 days for 3-month puts), and the strike nearest 5% below Nifty among strikes that traded at least 50 contracts that day. It is held to its real expiry and settled on the Nifty close.

Prices. A strike's own NSE closing price is used when at least 50 contracts traded that day. Otherwise the price comes from that day's implied-volatility smile, fitted to the liquid strikes of the same expiry. Single-contract prints can be 50% away from neighbouring strikes. 493 of 500 trades used the option's own price; the median put bought traded 1,510 contracts that day.

Like-for-like. The engine also prices the same real contracts with the course project's model. That separates the effect of real prices from the effect of real expiry dates. Period, portfolio, roll dates, thresholds and metrics are unchanged.

Limitations. Bhavcopies have end-of-day prices, not bid/ask quotes, so trading cost is an assumption. Positions are fractions of the portfolio, not whole lots. Dividends are left out. Six years contain one large crash, so the edge rests on few events.

Code, data download notebook and every number: output/results.json in the repository. For learning and research, not investment advice.